Generated by Scopus AI, Sat Oct 10 2026

Dual battery systems in dynamic wireless power transfer

Dual Battery Systems in Dynamic Wireless Power Transfer: Architectures, Coil Tuning, Magnetic Interference Mitigation, and Impact on Sensitive Components

Quick Reference

Key Findings Table

Topic	State-of-the-Art Solutions	Progress & Feasibility	Confidence Level	Gaps/Challenges
Dual Battery DWPT Architectures	Hybrid battery-supercapacitor systems, advanced control (PI-MPC, dual-phase shift), multi-transmitter/receiver topologies	Efficiencies >90%, robust to misalignment, real-time SoC optimization	High	Standardization, long-term field validation
Coil Tuning & Geometry	Auto-tuning, negative impedance converters, Tetris/Double-D/vertical coils, ferrite/nanocrystalline cores	Quality factor ↑, misalignment tolerance ↑, efficiency up to 99%	High	Trade-off: efficiency vs. shielding loss
Magnetic Interference & Eddy Currents	Passive (ferrite, litz wire, compensation loops), active (shielding coils), composite/hybrid shields	EMI reduction >80%, compliance with ICNIRP/ISO, minimal efficiency loss	High	Complexity, cost, dynamic adaptability
Sensitive Component Protection	Passive/active/composite shielding, sensor calibration, placement optimization	EMI reduction >70%, sensor error <2%, robust calibration methods	High	Real-world deployment, dynamic calibration
System Design Pathway	Integrated FEM simulation, iterative control/energy management, concurrent shielding/sensor validation	Validated in lab, modular, scalable	High	Field deployment, interoperability

Direct Answer

Current dual battery DWPT solutions employ hybrid energy management (battery + supercapacitor), advanced control (PI-MPC, dual-phase shift), and optimized coil tuning (auto-tuning, negative impedance converters, advanced geometries). Efficiencies above 90% are routinely achieved, with robust EMI mitigation via passive and

active shielding. To protect sensitive units (accelerometers, odometers), integrate both passive (ferrite/litz) and active (shielding coils) methods, validated through simulation and experiment. A recommended pathway: (1) FEM-based coil and shield design, (2) iterative control/energy management testing, (3) concurrent EMI/sensor validation. These approaches are strongly supported by recent literature and meta-analyses [1] [2] [3] [4].

Study Scope

- **Time Period:** 2018–2024
- **Disciplines:** Power electronics, automotive engineering, electromagnetic compatibility, sensor systems
- **Methods:** Meta-analysis, empirical studies, simulation (FEM), experimental validation, optimization algorithms

Assumptions & Limitations

- Most findings are validated in laboratory or controlled testbeds; real-world, long-term deployment data is limited.
- Standardization and interoperability across manufacturers remain open challenges.
- Sensor interference studies focus on typical automotive sensors; extreme environments may require further validation.

Suggested Further Research

- Long-term field studies on battery aging and EMI effects in real traffic.
- Standardized protocols for control communication and sensor calibration.
- AI-driven adaptive control for real-time coil/shield tuning and energy management.
- Holistic integration of EMI mitigation with vehicle communication and safety systems.

1. Introduction

Overview of Dynamic Wireless Power Transfer and Dual Battery Systems

Dynamic wireless power transfer (DWPT) enables electric vehicles (EVs) to charge while in motion, addressing range anxiety and reducing the need for large, heavy batteries. Dual battery systems—often combining batteries with supercapacitors—are increasingly integrated to buffer power fluctuations, optimize state-of-charge (SoC), and extend battery life. However, DWPT introduces challenges: fluctuating coupling, misalignment, electromagnetic interference (EMI), and the need to protect sensitive components (e.g., accelerometers, odometers) from magnetic disturbances. This report synthesizes the latest solutions, progress, and design pathways for dual battery DWPT systems, with a focus on coil tuning, magnetic interference mitigation, and sensor protection [5] [6].

2. Current Solutions and Progress in Dual Battery DWPT Systems

2.1 Technological Advancements and Architectures

- **Hybrid Architectures:** Integration of batteries and supercapacitors in DWPT receivers smooths power delivery, buffers against coupling fluctuations, and extends battery life [7] [8].
- **Advanced Control:** Dual-phase shift, extended phase shift modulation, and PI-MPC controllers dynamically adjust power flow, maintaining efficiency under misalignment and dynamic coupling [9] [10].
- **Multi-Transmitter/Receiver Topologies:** Multi-leg converters and multi-coil arrays enable constant charging and robust operation under misalignment [11] [12].
- **Voltage Balancing Circuits:** Reconfigurable multiport charging and IPT-based voltage equalizers maintain robust voltage balance for battery/supercapacitor banks, even under misalignment [13] [14].

Synthesis: These architectures achieve high efficiency (>90%), robust power delivery, and improved battery longevity, but require careful control and system integration [9] [10].

2.2 Dual Battery Management and Energy Optimization

- **Hybrid Energy Management:** Offline-online hybrid strategies, neural network-based SoC estimation, and hierarchical real-time control optimize SoC, reduce battery aging, and adapt to dynamic driving [15] [16].
- **Supercapacitor Integration:** Supercapacitors buffer transient loads, stabilize output, and reduce battery stress, but require advanced impedance-aware control to manage dynamic reflected impedance [17] [18].
- **Optimization Algorithms:** Genetic algorithms, reinforcement learning, and fuzzy logic controllers are used for multi-objective optimization of energy flow, battery health, and operational cost [19] [20].

Synthesis: These methods enable precise, adaptive energy management, but computational complexity and real-world adaptability remain challenges [20].

2.3 Control Delays, Interoperability, and Scalability

- **Control Delays:** Real-time control delays (e.g., communication, computation) increase output ripple and degrade performance; model predictive control with delay compensation mitigates these effects [21].
- **Interoperability:** Decoupling self-adaptive receivers and modular architectures address compatibility across different DWPT configurations [12] [22].
- **Scalability:** Multi-receiver LCC-S topologies and modular platforms (e.g., DEXTER) enable near-linear power scaling and facilitate system expansion [12] [23].

Synthesis: While advanced control and modularity improve scalability and interoperability, standardization and robust communication infrastructure are needed for large-scale deployment [24].

3. Coil Tuning Methods for Wireless Power Transfer

3.1 Inductive and AC Coil Tuning Techniques

- **Negative Impedance Converters (NIC):** NICs drastically increase coil quality factor (Q), reduce losses, and boost efficiency (up to 96%) [25] [26].
- **Auto-Tuning Algorithms:** Systems leveraging self-inductance variation and auto-tuning control maintain resonance and efficiency without complex feedback, achieving up to 96.7% efficiency [27].
- **Dual-Loop PI/MPC Control:** Ensures stable power delivery under time-varying coupling and misalignment [28].

Synthesis: These methods enable robust, high-efficiency power transfer in dynamic environments, with minimal manual intervention [26] [27].

3.2 Coil Geometry and Material Optimization

- **Advanced Geometries:** Tetris-shaped, Double-D, and vertical receiver coils with ferrite cores improve misalignment tolerance and efficiency (up to 99%) [29] [30].
- **Material Selection:** Ferrite, nanocrystalline, and litz wire cores optimize coupling, reduce losses, and enhance EMI suppression [31] [32].
- **Genetic Algorithm Optimization:** Used for multi-coil placement and geometry, maximizing efficiency under dynamic conditions [33] [34].

Synthesis: Optimized coil design is critical for balancing efficiency, misalignment tolerance, and EMI mitigation [30] [31].

3.3 Frequency Tuning and Resonance Maintenance

- **Frequency Tracking:** Variable-frequency control and segmented frequency-tracking maintain resonance and efficiency over large misalignments/distances (>75% efficiency up to 24 cm) [35] [36].
- **Compensation Network Optimization:** Fine-tuning compensation capacitors and inductors adapts to load/coupling changes, sustaining high efficiency [37] [38].

Synthesis: Dynamic frequency tuning is essential for robust DWPT operation, especially under real-world misalignment and load variation [35] [36].

4. Magnetic Interference and Eddy Current Mitigation

4.1 Passive and Active Magnetic Shielding

- **Passive Compensation Loops:** Simple, robust, and cost-effective; reduce magnetic emissions by up to 42%, but require careful design to avoid negative coupling effects [3] [39].
- **Active Shielding Coils:** Independently powered coils reduce EMI by >27 dB with minimal efficiency loss (<1%), effective for compliance with safety standards [40] [41].
- **Hybrid Approaches:** Combining passive and active methods yields superior shielding with only slight efficiency reduction [42] [43].

Synthesis: Both passive and active shielding are effective; hybrid designs offer the best trade-off between EMI reduction and system efficiency [40] [43].

4.2 Eddy Current Suppression and Magnetic Structure Optimization

- **Ferrite/Nanocrystalline Cores:** Diagonal-laminated and flexible nanocrystalline cores reduce eddy current losses, improve efficiency (>93%), and lower weight/volume [44] [45].
- **Segmentation/Lamination:** Interrupts eddy current paths, reducing heat generation and power losses [46] [47].
- **Material Selection:** Low-conductivity, high-permeability materials (e.g., nanocrystalline alloys) minimize losses across frequencies [48] [49].

Synthesis: Optimized core materials and structures are essential for minimizing eddy current losses and maintaining high DWPT efficiency [44] [46].

4.3 Compensation Network Tuning and Trade-offs

- **Resonance Capacitance Selection:** Reduces leakage fields by up to 42% without affecting input impedance or efficiency [39] [50].
- **Double-Sided LCC Compensation:** Enables fine control of emissions and maintains high efficiency under load/coupling variation [50] [51].
- **Multi-Objective Optimization:** Balances EMI reduction, efficiency, and system robustness [52].

Synthesis: Careful compensation network design is critical for EMI mitigation without sacrificing power transfer performance [39] [50].

5. Magnetic Shielding Techniques for Sensitive Component Protection

5.1 Passive vs Active Shielding: Comparative Analysis

- **Passive Shields (Ferrite, Litz Wire):** Reduce stray fields by up to 75%, with minimal efficiency loss if optimized; litz wire shields are lighter and more effective than aluminum [53] [54].
- **Active Shields:** Achieve higher field reduction (up to 27 dB), with negligible efficiency impact; require additional power/control [41] [55].
- **Hybrid Designs:** Combine passive and active elements for >85% field reduction with <1% efficiency loss [43] [56].

Synthesis: Hybrid shielding is recommended for critical sensor protection, balancing effectiveness and efficiency [43] [56].

5.2 Composite and Hybrid Shielding Structures

- **Composite Shields:** Integrate ferrite, aluminum, and active coils; reduce material usage by >40% while maintaining safety and efficiency [53] [57].
- **Optimization Algorithms:** Genetic algorithms and multi-objective optimization minimize material use and maximize shielding [58] [59].

Synthesis: Composite/hybrid shields are optimal for automotive applications, offering material and cost savings with robust EMI mitigation [57] [58].

5.3 Optimization and Placement Strategies

- **Shield Placement:** Strategic positioning of shields and coils maximizes field reduction in sensitive zones without degrading DWPT performance [42] [60].
- **Algorithmic Optimization:** Particle swarm, genetic, and gradient-based algorithms optimize shield geometry and coil placement [40] [60].

Synthesis: Algorithmic optimization and careful placement are essential for effective sensor protection in complex vehicle environments [40] [60].

6. Impact of DWPT Magnetic Fields on Sensitive Components

6.1 Quantified Effects on Sensor Accuracy

- **Sensor Error:** Magnetic disturbances can cause up to 27% variation in sensor readings; optimized shielding and placement reduce errors to <2% [61] [62].
- **Misalignment Effects:** Lateral/vertical misalignment estimation errors are typically <2 cm; advanced calibration and sensor fusion further reduce errors [63] [64].

Synthesis: With proper shielding and calibration, sensor accuracy can be maintained within automotive requirements [62] [63].

6.2 Magnetic Shielding Materials and Configurations

- **Material Selection:** MnZn ferrite, nanocrystalline alloys, and high-entropy alloys offer superior EMI suppression and mechanical robustness [65] [66].
- **Shield Geometry:** Circular closed-shields and double-layer passive coils maximize attenuation [67] [68].

Synthesis: Material and geometry selection are critical for effective EMI mitigation in automotive sensors [65] [68].

6.3 Active Shielding and Cancellation Loop Effectiveness

- **Active Coils:** Reduce magnetic fields by up to 39.5 dB with minimal efficiency loss; effective for compliance and sensor protection [40] [41].
- **Cancellation Loops:** Require precise tuning; can achieve up to 99% field reduction for encircled circuits [69] [70] .

Synthesis: Active methods are highly effective, especially when combined with passive shielding [41] [69].

6.4 Sensor Calibration and Placement Optimization

- **Calibration Methods:** Gravity-based, least squares, and evolutionary algorithm-based calibration reduce sensor errors without extra hardware [71] [72].
- **Placement Optimization:** Particle swarm and multi-magnetometer fusion algorithms minimize measurement errors due to interference [73] [74].

Synthesis: Regular calibration and optimized placement are essential for reliable sensor operation in DWPT environments [72] [74].

7. Pathway for System Design and Knowledge Acquisition

7.1 Integrated System Design Recommendations

- **Step 1:** FEM-based simulation for coil, shield, and compensation network design (optimize geometry, material, and placement) [28] [75].
- **Step 2:** Iterative testing of auto-tuning and dual-loop PI/MPC controllers under dynamic coupling/misalignment [9] [28].

- **Step 3:** Integration of hybrid energy management (neural network-based SoC estimation, supercapacitor buffering) [7] [16].
- **Step 4:** Concurrent implementation of passive/active shielding and eddy current suppression; validate with sensor calibration and EMI testing [67] [76].
- **Step 5:** Modular, scalable architecture for multi-device and multi-receiver expansion [12] [23].

Synthesis: A holistic, simulation-driven, and modular approach ensures robust, efficient, and EMI-compliant DWPT system design [9] [28] [75] [76].

7.2 Knowledge Acquisition on Eddy Currents and Magnetic Shielding

- **Eddy Currents:** Arise from time-varying magnetic fields in conductive materials; suppressed via core segmentation, lamination, and material selection (nanocrystalline, ferrite) [44] [46].
- **Magnetic Shielding:** Achieved through passive (ferrite, litz wire, aluminum), active (powered coils), and composite/hybrid structures; optimized via simulation and genetic algorithms [67] [77].
- **Learning Resources:** FEM simulation tools, optimization algorithm libraries, and recent review articles on DWPT EMI mitigation [44] [77].

Synthesis: Mastery of electromagnetic simulation, material science, and optimization is essential for effective DWPT system design and sensor protection [44] [77].

7.3 Future Research Directions

- **Field Validation:** Long-term, real-world studies on battery aging, EMI effects, and sensor reliability [78] [79].
- **Standardization:** Development of interoperable protocols for control, communication, and calibration [78].
- **AI-Driven Control:** Adaptive, closed-loop systems for real-time coil/shield tuning and energy management [80] [81].
- **Holistic Integration:** Unified frameworks for power, communication, and EMI mitigation in automotive platforms [79].

Synthesis: Addressing these gaps will enable robust, scalable, and safe deployment of dual battery DWPT systems in real-world automotive environments [78] [79] [80] [81].

Conclusion

Recent advances in dual battery DWPT systems—spanning hybrid energy management, advanced coil tuning, and robust EMI mitigation—enable high-efficiency, scalable, and sensor-safe wireless charging for electric vehicles. While laboratory results are promising, further work is needed to standardize protocols, validate long-term

performance, and integrate adaptive, AI-driven control for real-world deployment. A holistic, simulation-driven, and modular design pathway, coupled with rigorous sensor protection strategies, is recommended for researchers and practitioners advancing the state of DWPT technology [1] [2] [3] [4].

References

1. Comparative Analysis of DWPT Topologies and Regulation Schemes for Improved Controllability Bagchi, A.C., Kamineni, A., Zane, R., Carlson, R.B. ECCE 2020 - IEEE Energy Conversion Congress and Exposition, 2020 https://www.scopus.com/pages/publications/85097200806?origin=scopusAI

2. Designing efficient coils for a 500W Inductive Wireless Power Transfer (WPT) system using Finite Element Method Magnetics (FEMM) Phuangchaosuan, S., Kamnarn, U., Chaoumead, A., (...), Phaengkieo, D. 2026 International Power Electronics Conference, IPEC-Nagasaki 2026 - ECCE Asia, 2026 https://www.scopus.com/pages/publications/105046056585?origin=scopusAI

3. Mitigation of Magnetic Fields in Dynamic Wireless Power Transfer Systems through Passive Compensation Loops Campi, T., Cruciani, S., Maradei, F., Feliziani, M. Proceedings of 2024 IEEE Wireless Power Technology Conference and Expo, WPTCE 2024, 2024 https://www.scopus.com/pages/publications/85197430626?origin=scopusAI

4. Analysing the impact of magnetic coupler materials for wireless power transfer in electric vehicles Yadav, A., Bera, T.K. International Journal of Vehicle Design, 2025 https://www.scopus.com/pages/publications/105014392203?origin=scopusAI

5. Review and Comparative Analysis of Topologies and Control Methods in Dynamic Wireless Charging of Electric Vehicles Bagchi, A.C., Kamineni, A., Zane, R.A., Carlson, R. IEEE Journal of Emerging and Selected Topics in Power Electronics, 2021 https://www.scopus.com/pages/publications/85101428667?origin=scopusAI

6. A Comprehensive Review of the On-Road Wireless Charging System for E-Mobility Applications Yuvaraja, S., Narayanamoorthi, R., Sathik Mohamed Ali, J., Almakhles, D. Frontiers in Energy Research, 2022 https://www.scopus.com/pages/publications/85134428564?origin=scopusAI

7. Hybrid Efficient Fast Charging Strategy for WPT Systems: Memetic-Optimized Control with Pulsed/Multi-Stage Current Modes and Neural Network SOC Estimation El Ancary, M., Lassioui, A., El Fadil, H., (...), Chiheb, M. World Electric Vehicle Journal, 2025 https://www.scopus.com/pages/publications/105011650641?origin=scopusAI

8. Hybrid method based energy management of electric vehicles using battery-super capacitor energy storage AlKawak, O.A., Kumar, J.R.R., Daniel, S.S., Reddy, C.V.K. Journal of Energy Storage, 2024 https://www.scopus.com/pages/publications/85179890279?origin=scopusAI

9. Smart wireless BMS for electric vehicles: role of advanced battery materials and energy-efficient communication technologies Awaar, V.K., Praveen, J., Ruchika, A., (...), Venkaiah, M. Journal of Physics: Conference Series, 2026 https://www.scopus.com/pages/publications/105035989979?origin=scopusAI

10. Hierarchical real-time energy management strategy of eVTOL hybrid power system based on finite-set Li, X., Zhang, H., Liu, C., (...), Li, P. Aerospace Science and Technology, 2026 https://www.scopus.com/pages/publications/105026131130?origin=scopusAI

11. A Five Leg Converter with Multi-Transmitter for an In-Motion Charging System Yuvaraja, S., Narayanamoorthi, R. Journal of Physics: Conference Series, 2022 https://www.scopus.com/pages/publications/85139093412?origin=scopusAI

12. Analysis of Wide Range CV DWPT Systems With Multi-Receiver Architecture Featuring Mobile-Powered Electronics Li, J., Zhu, C., Ji, J., (...), Zhang, X. IEEE Transactions on Consumer Electronics, 2026 https://www.scopus.com/pages/publications/105034755352?origin=scopusAI

13. Multi-State Voltage Balancing of UAV's Cell String: A Reconfigurable WPT Based Multiport Hybrid Charging Approach Cai, C., Wang, J., Luo, Y., (...), Liu, L. IEEE Transactions on Industrial Electronics, 2025 https://www.scopus.com/pages/publications/86000328300?origin=scopusAI

14. A Compact IPT-Based Voltage Equalizer With a Symmetrical Interleaved Coil for Robust Drone Battery Balancing Under Misalignment Liu, L., Li, Y., Yu, X., (...), Xie, K. IEEE Transactions on Industrial Electronics, 2026 https://www.scopus.com/pages/publications/105034164442?origin=scopusAI

15. Research on charging behavior and energy management strategy of electric vehicles in dynamic wireless power transfer scenarios Li, A., Guo, Z., Duan, S., Zhang, Y. Journal of Energy Storage, 2026 https://www.scopus.com/pages/publications/105046325018?origin=scopusAI

16. Physics-Informed Multi-Head Attention-Based SOC Estimation for LFP Batteries Using Only Measured Signals Sorouri, H., Oshnoei, A., Marinescu, M., Teodorescu, R. IEEE Access, 2026 https://www.scopus.com/pages/publications/105036557470?origin=scopusAI

17. Supercapacitor Assisted PV Powered Dynamic Wireless Power Charging System for EV Nayak, U., Pati, A.K. 2024 4th International Conference on Intelligent Technologies, CONIT 2024, 2024 https://www.scopus.com/pages/publications/85202756984?origin=scopusAI

18. Wireless power transfer systems with supercapacitor energy management: A comprehensive review Liu, Z., Silva Thotabaddadurage, S., Lie, T.T. Journal of Energy Storage, 2026 https://www.scopus.com/pages/publications/105051208966?origin=scopusAI

19. A Novelty of Several Optimization Criteria Hybrid Energy Storage System and Power Management in EV Applications Sankarananth, S., Santhosh Kumar, G., Kavitha, G. E3S Web of Conferences, 2023 https://www.scopus.com/pages/publications/85170087313?origin=scopusAI

20. Comprehensive Research on Advanced Energy Management Strategies for Hybrid Energy Storage Systems in Electric Vehicles Parvathi, Asundi, A.K., Sujatha, D., (...), Rajashekar, K. 4th International Conference on Integrated Circuits and Communication Systems, ICICACS 2026, 2026 https://www.scopus.com/pages/publications/105044451973?origin=scopusAI

21. Skew-Robust Model Predictive Control for Dynamic Wireless Power Transfer System of Electric Vehicles Under Multisource Delays Li, J., Zhou, Z., Dai, Y., (...), Chen, Q. IEEE Transactions on Transportation Electrification, 2026 https://www.scopus.com/pages/publications/105039272478?origin=scopusAI

22. An Interoperable Dynamic Wireless Charging System with Stable Output Based on a Self-Adaptive Two-Pole Receiver Zhang, Y., Zhou, H., Shen, Z., (...), Mao, X. IEEE Transactions on Power Electronics, 2024 https://www.scopus.com/pages/publications/85195412101?origin=scopusAI

23. Toward Sustainable E-Mobility: Optimizing the Design of Dynamic Wireless Charging Systems Through the DEXTER Experimental Platform Di Capua, G., Femia, N., Maffucci, A., (...), Fontana, N. Sustainability (Switzerland), 2026 https://www.scopus.com/pages/publications/105035623899?origin=scopusAI

24. Communication infrastructure for dynamic wireless charging of electric vehicles Echols, A., Mukherjee, S., Mickelsen, M., Pantic, Z. IEEE Wireless Communications and Networking Conference, WCNC, 2017 https://www.scopus.com/pages/publications/85019693274?origin=scopusAI

25. High Q-factor WPT system with negative impedance converter Yoon, S.-H., Kim, T.-H., Yook, J.-G., (...), Lee, W.-Y. WPTC 2017 - Wireless Power Transfer Conference, 2017 https://www.scopus.com/pages/publications/85024364790?origin=scopusAI

26. High Q-Factor Coil with Transistorized Negative Impedance Converter for Mobile Applications Kim, T.-H., Yun, G.-H., Yook, J.-G. 2019 IEEE Wireless Power Transfer Conference, WPTC 2019, 2019

https://www.scopus.com/pages/publications/85083572479?origin=scopusAI

27. Automatic Current Control by Self-Inductance Variation for Dynamic Wireless EV Charging Jeong, S.Y., Park, J.H., Hong, G.P., Rim, C.T. 2018 IEEE PELS Workshop on Emerging Technologies: Wireless Power Transfer, Wow 2018, 2018 https://www.scopus.com/pages/publications/85053893472?origin=scopusAI

28. Dual-Loop PI Control for a Dynamic Wireless Electric Vehicle Charging System with Dual-transmitters using 3D Modeling of DD Coils Hassan, A.U., Aula, F.T. Journal of Visualized Experiments, 2026 https://www.scopus.com/pages/publications/105039934394?origin=scopusAI

29. Vertical Receiver Coils for Improved Energy Transfer Efficiency in Dynamic Wireless Charging of Electric Vehicles Garrofy, I., Roitman, A., Nikulshin, Y., Wolfus, S. Applied Sciences (Switzerland), 2025 https://www.scopus.com/pages/publications/105002276575?origin=scopusAI

30. Research on misalignment tolerance of concave planar spiral coils for wireless power transfer systems Yi, Y., Zhihao, L., Lu, Z., (...), Zhaoyi, Z. Yi Qi Yi Biao Xue Bao/Chinese Journal of Scientific Instrument, 2025 https://www.scopus.com/pages/publications/105036477757?origin=scopusAI

31. Misalignment Tolerance of Inductive Power Transfer Coupler with Low Loss and High Magnetic Induction Ferromagnetic Materials Kang, J., Cheng, S., Wang, Y., (...), Zhao, H. Conference Record - IAS Annual Meeting (IEEE Industry Applications Society), 2022 https://www.scopus.com/pages/publications/85142840646?origin=scopusAI

32. Performance Analysis of a High-efficiency Multi-winding Wireless EV Charging System Using U-U and U-I Core Geometries Pathipati, V.K., Azeez, N.A., Aditya, K., (...), Botting, C. 2016 IEEE Transportation Electrification Conference and Expo, ITEC 2016, 2016 https://www.scopus.com/pages/publications/84994339024?origin=scopusAI

33. Efficiency optimization for multi-coil-domino wireless power transfer system Zhu, H., Deng, W., Gong, C., (...), Lin, D. 2024 6th International Conference on Energy, Power and Grid, ICEPG 2024, 2024 https://www.scopus.com/pages/publications/85215108340?origin=scopusAI

34. Three-Dimensional Free-Positioning Wireless Power Transfer via Multiple-Current Amplitude Modulation Lyu, J., Liu, W., Niu, S., (...), Chau, K.T. IEEE Journal of Emerging and Selected Topics in Power Electronics, 2025 https://www.scopus.com/pages/publications/85219134969?origin=scopusAI

35. A frequency-tuned magnetic resonance-based wireless power transfer system with near-constant efficiency up to 24 cm distance Sis, S.A., Kavut, S. Turkish Journal of Electrical Engineering and Computer Sciences, 2018 https://www.scopus.com/pages/publications/85063013110?origin=scopusAI

36. Study on Frequency Splitting and Segmented Tracking for High-Performance WPT under Horizontal Offset Li, J., Zou, J., Zou, Y., Li, Z. Progress in Electromagnetics Research C, 2026 https://www.scopus.com/pages/publications/105038078539?origin=scopusAI

37. An Optimal Variable-Frequency Asymmetrical-Phase-Shift Strategy for Wireless Power Transfer System Chen, H., Chen, C., Jia, S., (...), Xu, Z. Proceedings of 2022 IEEE 5th International Electrical and Energy Conference, CIEEC 2022, 2022 https://www.scopus.com/pages/publications/85137331340?origin=scopusAI

38. Automatic Position-Tracking and Power-Allocating of Dynamic Wireless Power Transfer With Ferrite Shielding Layers Dai, Z., Tang, Y., Xiao, X., (...), Fang, Z. IEEE Transactions on Transportation Electrification, 2025 https://www.scopus.com/pages/publications/85192960030?origin=scopusAI

39. Resonance Capacitance Selection Method for Minimizing Leakage Magnetic Fields and Achieving Zero Phase Angles in Wireless Power Transfer Systems Shin, Y., Rhee, J., Woo, S. Electronics (Switzerland), 2024 https://www.scopus.com/pages/publications/85208441850?origin=scopusAI

40. Active shielding design and optimization of a wireless power transfer (WPT) system for automotive Cruciani, S., Campi, T., Maradei, F., Feliziani, M. Energies, 2020 https://www.scopus.com/pages/publications/85094944135?origin=scopusAI

41. Analysis of Wireless Power Transfer System Employing Active Shielding with Virtual Inductance and Two-port Equivalent Circuit Furukawa, K., Kusaka, K., Itoh, J.-I. 2021 IEEE Energy Conversion Congress and Exposition, ECCE 2021 - Proceedings, 2021 https://www.scopus.com/pages/publications/85123374872?origin=scopusAI

42. Electromagnetic shielding design for magnetic coupler of N-type dynamic electric vehicle wireless power transfer systems Wang, S., Wei, B., Wu, X., (...), Xu, J. 2019 22nd International Conference on Electrical Machines and Systems, ICEMS 2019, 2019 https://www.scopus.com/pages/publications/85077132585?origin=scopusAI

43. Electromagnetic Shielding Techniques in the Wireless Power Transfer System for Charging Inspection Robot Application Jiao, C., Xu, Y., Li, X., (...), Pang, C. International Journal of Antennas and Propagation, 2021 https://www.scopus.com/pages/publications/85111036858?origin=scopusAI

44. A Novel Diagonal-Laminated Fe-Based Nanocrystalline Ribbon Core Structure in Wireless Power Transfer Systems Wang, D., Cui, S., Zhang, J., (...), Zhu, C. IEEE Student Conference on Electric Machines and Systems (SCEMS), 2024 https://www.scopus.com/pages/publications/85212300730?origin=scopusAI

45. Research on Novel Flexible High-Saturation Nanocrystalline Cores for Wireless Charging Systems of Electric Vehicles Xiong, M., Wei, X., Huang, Y., (...), Dai, H. IEEE Transactions on Industrial Electronics, 2021 https://www.scopus.com/pages/publications/85112292372?origin=scopusAI

46. Reducing the losses in sintered permalloy by addition of ferrite Plaza, M., Pérez, L., Sánchez, M.C. Journal of Magnetism and Magnetic Materials, 2007 https://www.scopus.com/pages/publications/33750509406?origin=scopusAI

47. Control of eddy current heating in magnetic flux conductors under alternating magnetic field Xiaojian, Y., Peng, C., Qiming, C., (...), Wenbo, Z. Harbin Gongye Daxue Xuebao/Journal of Harbin Institute of Technology, 2026 https://www.scopus.com/pages/publications/105049611694?origin=scopusAI

48. Efficiency Characteristics of Loosely Coupled Transformer With Laminated Nanocrystalline Core Considering Nonlinear Factors Caused by Crushing and Compressing Process Li, L., Chen, J., Zhang, X., (...), Zhao, H. IEEE Transactions on Industry Applications, 2024 https://www.scopus.com/pages/publications/85199041872?origin=scopusAI

49. Novel magnetic core materials impact modelling and analysis for minimization of RF heating loss Ghosh, B.K., Mohamad, K.A., Saad, I. AIP Conference Proceedings, 2016 https://www.scopus.com/pages/publications/84984549423?origin=scopusAI

50. Magnetic Field Mitigation in Dynamic Wireless Power Transfer Systems by Double Sided LCC Compensation Campi, T., Cruciani, S., Maradei, F., Feliziani, M. IEEE Access, 2024 https://www.scopus.com/pages/publications/85200814599?origin=scopusAI

51. A Double-Sided LCC Compensation Network and Its Tuning Method for Wireless Power Transfer Li, S., Li, W., Deng, J., (...), Mi, C.C. IEEE Transactions on Vehicular Technology, 2015 https://www.scopus.com/pages/publications/84933055852?origin=scopusAI

52. Research on multi-objective parameter optimization and switching control strategy of IPT system Huang, W., Hu, Y., Huang, J., (...), Chang, Y. Dianji yu Kongzhi Xuebao/Electric Machines and Control, 2024 https://www.scopus.com/pages/publications/85216117891?origin=scopusAI

53. Research on Single-Hole Compensated Passive Magnetic Shielding Structure for Electric Vehicle Wireless Power Transfer Systems Li, Z., Gan, Z., Ren, L., (...), Li, J. Progress In Electromagnetics Research B, 2024

https://www.scopus.com/pages/publications/85199577255?origin=scopusAI

54. Comparison of passive shields for coils in inductive power transfer Lu, M., Ngo, K.D.T. Conference Proceedings - IEEE Applied Power Electronics Conference and Exposition - APEC, 2017 https://www.scopus.com/pages/publications/85019982796?origin=scopusAI

55. Enhancing shielding efficiency of separate shell magnetic shield using two independent controllers Mahgoub, A., Sasada, I. IEEE Transactions on Magnetics, 2014 https://www.scopus.com/pages/publications/84957928229?origin=scopusAI

56. Surrounding Active Shielding Coil Design and Efficient Multiobjective Optimization for WPT Yu, Q., Lv, G., Pi, S., (...), Wang, T. IEEE Transactions on Transportation Electrification, 2025 https://www.scopus.com/pages/publications/105000867733?origin=scopusAI

57. Study on Composite Structure of Tian-Font Magnetic Shielding and Anti-Series Active Coils for Wireless Power Transfer System Li, Z., Zhang, W., Gan, Z., Li, B. CPSS Transactions on Power Electronics and Applications, 2025 https://www.scopus.com/pages/publications/105013543954?origin=scopusAI

58. Application of genetic algorithm in the design of the electromagnetic shielding Fu, J.-W., Hou, C.-Z., Dou, L.-H. Xi Tong Gong Cheng Yu Dian Zi Ji Shu/Systems Engineering and Electronics, 2004 https://www.scopus.com/pages/publications/84882717060?origin=scopusAI

59. Design of uniform field coils coupled with a magnetic shield cylinder Huang, B., Cao, Q., Chen, Z., (...), Zhai, Y. Journal of Magnetism and Magnetic Materials, 2023 https://www.scopus.com/pages/publications/85166658720?origin=scopusAI

60. Shielding Sensor Coil to Reduce the Leakage Magnetic Field and Detect the Receiver Position in Wireless Power Transfer System for Electric Vehicle Son, S., Woo, S., Kim, H., (...), Ahn, S. Energies, 2022 https://www.scopus.com/pages/publications/85127750707?origin=scopusAI

61. Magnetic field measurement for human exposure assessment nearwireless power transfer systems in Kilohertz and Megahertz Bands Ishihara, S., Onishi, T., Hirata, A. IEICE Transactions on Communications, 2015 https://www.scopus.com/pages/publications/84948762735?origin=scopusAI

62. EV Misalignment Estimation in DWPT Systems Utilizing the Roadside Charging Pads Tavakoli, R., Shabanian, T., Dede, E.M., (...), Pantic, Z. IEEE Transactions on Transportation Electrification, 2022 https://www.scopus.com/pages/publications/85113211188?origin=scopusAI

63. A Static Coil Misalignment Study Using Gaussian Curve Fitting: Foundations for Sensorless Dynamic WPT Yahaya, J.A.F., Sukriyllah, M., Khan, R.B.M., Susanti, E. ASM Science Journal, 2026 https://www.scopus.com/pages/publications/105042797609?origin=scopusAI

64. Robust Misalignment Calibration for Magnetometer Array Based on Characteristic Magnetic Signals Peng, D., Chen, Y., Li, X., (...), Luo, L. IEEE Transactions on Instrumentation and Measurement, 2026 https://www.scopus.com/pages/publications/105034453039?origin=scopusAI

65. Optimizing magnetic and electrical properties of MnZn ferrites via spatially engineered Ni distribution Wu, T., Wang, C., Ma, Z., (...), Sun, K. Journal of Advanced Ceramics, 2026 https://www.scopus.com/pages/publications/105030863630?origin=scopusAI

66. High-entropy alloys as magnetic shielding materials in low-frequency wireless power transmission Yeh, C.-H., Hu, C.-H., Wu, H.-T., (...), Shih, C.-F. Materials Today Advances, 2024 https://www.scopus.com/pages/publications/85207798314?origin=scopusAI

67. Low-Frequency Magnetic Field Mitigation Using Recyclable Sintered Pure-Iron Powder Compacts for Safer Power and Sensor Infrastructure Barge, V., Satpute, N., Kulkarni, M. SSRG International Journal of Mechanical

Engineering, 2026 https://www.scopus.com/pages/publications/105052207253?origin=scopusAI

68. Study of a Double-Layer Passive Magnetic Shielding System for Electric Vehicle WPT Zhang, X., Ren, L., Kong, P., (...), Li, Z. Progress In Electromagnetics Research C, 2023 https://www.scopus.com/pages/publications/85170564278?origin=scopusAI

69. Field Cancellation for Circuits Encircled by VHF Wireless Power Transfer Coils Zan, X., Avestruz, A.-T. IEEE Transactions on Power Electronics, 2023 https://www.scopus.com/pages/publications/85137577876?origin=scopusAI

70. Optimal Cancellation Loop Termination to Reduce the Magnetic Field in WPT Systems Cruciani, S., Campi, T., Maradei, F., Feliziani, M. IEEE Transactions on Electromagnetic Compatibility, 2025 https://www.scopus.com/pages/publications/85218919769?origin=scopusAI

71. Gravity-Based Calibration for In-Situ Acceleration Sensors Thiel, R., Fromm, R., Strakosch, F., (...), Derbel, F. Conference Record - IEEE Instrumentation and Measurement Technology Conference, 2025 https://www.scopus.com/pages/publications/105012162635?origin=scopusAI

72. Evolutionary algorithm based 9DOF sensor board calibration Sarcevic, P., Pletl, S., Kincses, Z. SISY 2014 - IEEE 12th International Symposium on Intelligent Systems and Informatics, Proceedings, 2014 https://www.scopus.com/pages/publications/84911164105?origin=scopusAI

73. Optimization of magnetic sensor arrays for current measurement based on swarm intelligence and D-optimality Zhang, Z., Di Rienzo, L. COMPEL - The International Journal for Computation and Mathematics in Electrical and Electronic Engineering, 2009 https://www.scopus.com/pages/publications/70350425820?origin=scopusAI

74. Distortion-robust distributed magnetometer for underwater pose estimation in confined UUVs Christensen, L., Gaudig, C., Kirchner, F. OCEANS 2015 - MTS/IEEE Washington, 2015 https://www.scopus.com/pages/publications/84963971919?origin=scopusAI

75. Modeling and application of hybrid multi permeability Fe-based flexible nanocrystalline laminated ribbon core structure in WPT systems Wang, D., Cui, S., Zhang, J., (...), Zhu, C. Wireless Power Transfer, 2025 https://www.scopus.com/pages/publications/105007564648?origin=scopusAI

76. Optimization Design of Active and Passive Hybrid Shielding for Electric Vehicle's Wireless Power Transfer System Wu, Y., Wang, T., Yu, Q., (...), Guan, S. Applied Computational Electromagnetics Society Journal, 2026 https://www.scopus.com/pages/publications/105038004988?origin=scopusAI

77. Optimized Ferrite Design for Wireless Charging of Electric Vehicles a Cost-Efficient Approach Ragul, R., Nakkeeran, R., Bharatiraja, C. 2025 IEEE 4th International Conference on Smart Technologies for Power, Energy and Control, STPEC 2025 - Conference Report, 2025 https://www.scopus.com/pages/publications/105040729344?origin=scopusAI

78. A Comprehensive State-of-The-Art Review of Wireless Power Transfer for Electric Vehicles: Misalignment, Speed Effects, and Topological Limitations in Dynamic Operation Piedrahita-Echavarria, E.Y., Holguin-Londono, M., Escobar-Mejia, A. 2026 16th International Conference on Power, Energy, and Electrical Engineering, CPEEE 2026, 2026 https://www.scopus.com/pages/publications/105041508892?origin=scopusAI

79. Applications of the signal space separation method Taulu, S., Simola, J., Kajola, M. IEEE Transactions on Signal Processing, 2005 https://www.scopus.com/pages/publications/27644509833?origin=scopusAI

80. A Review of Research Developments in Fluctuation Suppression for Dynamic Wireless Energy Transmission in Electric Vehicles Zhang, J., Li, X. 2025 10th International Seminar on Computer Technology, Mechanical and Electrical Engineering, ISCME 2025, 2025 https://www.scopus.com/pages/publications/105031695707?origin=scopusAI

81. Review of Application Development and Research of Electromagnetic Shielding in Magnetic Coupling Resonant Wireless Power Transmission System Dou, R., Zhang, X., Li, Y., (...), Yang, Q. Zhongguo Dianji Gongcheng Xuebao/Proceedings of the Chinese Society of Electrical Engineering, 2023
https://www.scopus.com/pages/publications/85171153231?origin=scopusAI